



Project Title: Explainable and Prerequisite-Aware Knowledge Gap Detection Using Machine Learning and Deep Knowledge Tracing for Personalized Learning

Abstract

Traditional online learning and assessment platforms evaluate student academic performance through static numerical grades and binary pass/fail quiz scores. However, these traditional metrics fail to capture latent cognitive knowledge gaps, problem-solving hesitation, hint dependency patterns, or prerequisite skill dependency failures.

This project proposes EduGap AI, an end-to-end explainable framework that combines Gradient Boosted Decision Trees and Deep Knowledge Tracing (DKT) to detect cognitive gaps and deliver personalized, session-aware remediation.

Utilizing 525,534 student interaction logs from the ASSISTments benchmark dataset across 20 mathematics skills, our feature engine extracts 7 zero-label-leakage interaction attributes (log_ms_response, hint_ratio, attempt_hint_sum, user_skill_hints, etc.).

Our top machine learning model (XGBoost Classifier) achieves 91.56% Accuracy & 0.9880 ROC-AUC, while PyTorch LSTM / DKT achieves 85.77% Accuracy.

The system integrates a dedicated /review portal featuring side-by-side color-coded answer comparison cards (Your Answer vs Correct Answer), interactive quick filtering, and single-click print-ready A4 PDF study guide downloads.

Methods and Materials

1. Dataset & Preprocessing Pipeline:

Ingested 525,534 student interaction logs from the ASSISTments benchmark dataset spanning 20 core mathematics skills. Applied median imputation for missing response times, mode encoding for tutor interaction modes, and logarithmic scaling: $\log_ms_response = \ln(1 + ms_first_response)$.

2. Behavioral Feature Engineering Matrix (7 Core Attributes):

- correct: Binary item correctness score (1.0 = Correct, 0.0 = Incorrect).
- attempt_count: Count of repeated problem submission attempts per item.
- hint_count & hint_ratio: Normalized hint reliance ratio = $\text{hint_count} / (\text{hint_total} + 1)$.
- attempt_hint_sum: Combined problem friction index = $\text{attempt_count} + \text{hint_count}$.
- log_ms_response & log_overlap_time: Log-scaled first response latency & total interaction time.
- user_skill_hints & user_skill_attempts: Longitudinal student persistence & historical struggle memory.

3. Target Class Quantization & Stratification:

Quantized the Knowledge Gap Index into 3 balanced target classes: High Gap (>50% error/hint friction), Moderate Gap (25-50% friction), and Low Gap (<=25% friction).

Discussion & Diagnostic Explainability

- Multi-Class Diagnostic Explainability: Evaluated models via One-vs-Rest (OvR) ROC-AUC curves (0.9880 XGBoost AUC, 0.9552 RF AUC, 0.9455 DT AUC) & Seaborn confusion matrices.
- Behavioral Friction Cues: Latency and hint reliance ratios accurately separate high knowledge gaps from low gaps with zero class imbalance.

ML vs Deep Learning Model Accuracy (%)



Introduction & Objectives

Problem Statement:

Conventional e-learning tools evaluate students purely on final quiz accuracy, overlooking critical behavioral interaction cues such as first-response latency, repeated submission attempts, and hint reliance ratios. Consequently, struggling students are assigned generic practice problems rather than targeted remediation aligned with their underlying cognitive gaps.

Key Project Objectives:

- 1. Zero-Leakage Feature Engine: Construct a 7-attribute behavioral interaction matrix capturing latency and hint reliance.
- 2. Machine Learning Benchmark: Train and evaluate Decision Tree (81.56%), Random Forest (82.77%), and XGBoost (91.56% Accuracy, 0.9880 ROC-AUC).
- 3. Deep Learning Benchmark: Train 1D-CNN (84.97%), Simple RNN (85.47%), and PyTorch LSTM / DKT (85.77% Accuracy).
- 4. Multi-Class Explainability: Validate performance via One-vs-Rest (OvR) ROC-AUC curves (0.9880 AUC) and Seaborn confusion matrices.
- 5. Dedicated /review Portal: Develop an interactive review interface with side-by-side comparison cards (Your Answer vs Correct Answer).
- 6. Session-Aware Remediation: Restrict recommendations to 10 attended topics with single-click print-ready A4 PDF academic study guide downloads.

Results & Model Benchmarks

Machine Learning vs Deep Learning Model Benchmark Summary:

Model Architecture	Accuracy	ROC-AUC / Type
Decision Tree Classifier	81.56%	0.9455 AUC
Random Forest Ensemble	82.77%	0.9552 AUC
XGBoost Classifier (TOP ML)	91.56%	0.9880 AUC
1D-CNN (Deep Learning)	84.97%	Deep Learning
Simple RNN (Deep Learning)	85.47%	Deep Learning
LSTM / DKT (Deep Learning)	85.77%	Deep Learning

[BOLD]Best Deployed Model Summary:

- Best Deployed Model: XGBoost Classifier with 91.56% Accuracy and 0.9880 ROC-AUC Score.
- Top Deep Learning Model: LSTM / DKT with 85.77% Accuracy for sequential knowledge tracing.
- Model Artifacts Saved: best_model.pkl, label_encoder.pkl, lstm_model.pth.

Conclusions & Novelties

Summary of Core Project Contributions:

- 1. Best Deployed Model: XGBoost Classifier achieves top performance with 91.56% Accuracy and 0.9880 ROC-AUC.
- 2. Deep Knowledge Tracing: LSTM / DKT model captures sequential interactions with 85.77% Accuracy.
- 3. Zero Label Leakage: Engineered 7-attribute behavioral interaction matrix capturing latency and hint reliance.
- 4. Multi-Class Explainability: Validated performance via OvR ROC-AUC curves (0.9880 AUC) & Seaborn confusion matrices.
- 5. Dedicated /review Portal: Side-by-side color-coded comparison cards (Your Answer vs Correct) with quick filtering.
- 6. Session Remediation: Attended-topic recommendations (10 topics) with single-click print-ready A4 PDF study guides.

Project Team

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References

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